Actual source taper and exact jet determination

November 9, 2026. Mathematical source `31c212f5a`.

Unsigned producer evidence throughout [pcc:unverified].

Result. The actual nonzero source grid nodes are now counted by exact two-adic valuation. Their source ceiling taper has enough total multiplicity for the repaired polynomial's doubled degree. Exact scaled source jets determine the actual cleared packed polynomial over every $\mathbb{Q}$-algebra, including nilpotent coefficient rings. This continues the preceding affine-contraction component and does not replace its versioned report.

Actual grid counts. Let $L = 2^a$ and $m \in \mathrm{Fin}(2^{a+1})$, with actual source point $t_m = m - 2^a \in \mathbb{Z}$. For each active $i < a$, an explicit quotient/remainder bijection counts the residue class $m \bmod 2^{i+1} = 2^i$. Since $2^{i+1} \mid L$, shifting by L preserves it. The integer lemma

$$t \bmod 2^{i+1} = 2^i \quad \Longleftrightarrow \quad t \neq 0 \text{ and } v_2(t) = i$$

therefore gives exactly 2^{a-i} actual nodes. Zero is excluded explicitly; the convention for valuation at zero is not used as a source level.

Actual ceiling and all-level multiplicity. The source ceiling $h = \lceil 3M/u \rceil$ has its budget $3M \leq hu$ derived. Define $q(0) = 0$ and $q(t) = h \max(0, u - v_2(t))$ otherwise. The Lean proof identifies its sum over individual actual grid nodes with the complete level sum

$$\sum_m q(t_m) = \sum_{i=0}^{u-1} 2^{a-i} h(u-i).$$

For $3 \leq u \leq a$ and $u \leq M$, this satisfies

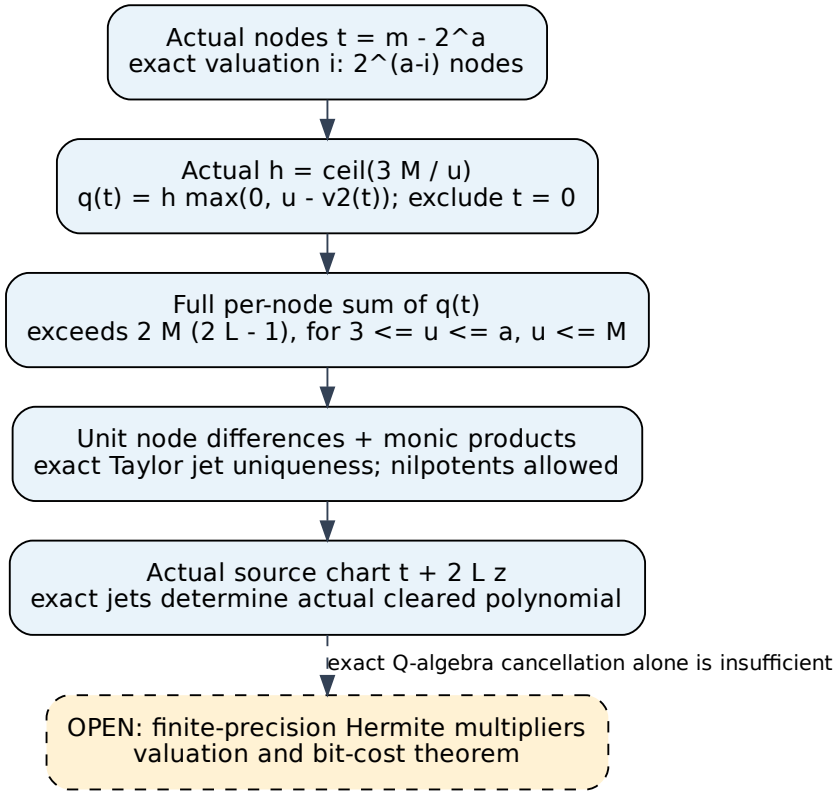
$$\sum_m q(t_m) > 2M(2L-1).$$

The first three actual levels certify a lower bound on this full sum. The source's ceiling and actual cardinalities are derived; no assumed level distribution or user-supplied budget replaces them. The odd-node lower bound alone would not justify the doubled degree.

Exact uniqueness. For nodes with unit pairwise differences over a commutative ring, the factors $(X - t_i)$ are pairwise coprime. Equality of exact Taylor coefficients below q_i implies $(X - t_i)^{q_i} \mid F - G$. Their monic product divides $F - G$ and has degree $\sum_i q_i$. If both polynomial degrees are below that sum, then $F - G = 0$. This proof permits zero divisors and nilpotents.

Distinct actual source nodes have nonzero rational differences, which map to units in any $\mathbb{Q}$-algebra. The actual chart is $t + 2Lz$; its coefficient of order j equals Taylor coefficient j times $(2L)^j$. This unit scaling connects actual scaled jets to the actual cleared packed polynomial's degree bound and gives the advertised determination theorem.

Precision boundary. Scaling cancellation uses the $\mathbb{Q}$-algebra, where $2L$ is a unit. It cannot be repeated modulo 2^p. This component proves exact determination; finite-precision reconstruction, construction costs, efficient evaluation of the contraction and a recursive bit-cost bound remain separate obligations.



Solid boxes describe this component; the dashed box remains open.

Checks. Both modules are default targets. The default build passed, and the actual source-discovering gate reports 162 declarations with zero extra trusted axioms.

Foundations remain `propext`, `Classical.choice`, `Quot.sound`. The import-inclusive fresh replay checks `HermiteUniqueness`. Its status is in `lean/recovery-fresh-status.json`, with the raw log beside it. All checks remain unsigned producer evidence [pcc:unverified].

Pinned toolchain: Lean 4.35.0. CryptoLean: `148f5dc389b341401e06d76abe7783c4ccf42f2e`. Mathlib: `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`. No checker, proof gate, kernel, trust policy or dependency revision changed. Failed node/API elaborations and both passing Hermite builds are retained. No timing or performance inference is drawn from compiler logs. Earlier plots and scoreboards remain historical because no measured value changed.

Proof map and unsigned records. Canonical statement hashes are in `lean/recovery-claim-pins.json`.

Actual nonzero level count	`NodeMultiplicity.actualValuationCard` PCC-20261109-ed63ef
Actual ceiling/per-node margin	`NodeMultiplicity.sourceJetDegreeMargin` PCC-20261109-966d63
Exact ring-valued uniqueness	`HermiteUniqueness.exactHermiteUnique` PCC-20261109-7e8217

Actual packed scaled jets	`HermiteUniqueness.`
	`actualPackedScaledJetsDetermine`
	PCC-20261109-05e9da

The identities involve finite counted sets, valuations and variable-length jet vectors. They do not fit polynomial-expression certificate schema v1; their evidence is the quantified Lean source and source audit. The preceding affine polynomial certificate is unchanged.

Sources and scope. Reviewed OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026, D.4.3–D.4.4 and D.8–D.10, plus the repository's `matched-recovery-independent.md` and actual Mathlib remainder, ceiling, Taylor, coprime-product and monic-degree APIs. Primary PDF SHA256: `5b1612dc2eebb2cad5b4cbe62d1fed2982f01ab1908c26f70010a651487e8123`. Pinned source: https://github.com/openai/math/blob/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026/paper.pdf. The KB MCP is absent and repository fallback unavailable (exit 127). This is a bounded source review and formalization of established interpolation, with no novelty or exhaustion claim. No curve or subgroup solve was attempted.

Next discriminating proof. Formalize the actual normalized Hermite basis E_t from D.9 and bound its multiplier valuations. D.10 uses the weighted level difference with factor $j/2^j$; omitting j gives the wrong series. Its taper inequality yields $v_2(E_t(0)) \geq -6hD_i$ via $\sum_{j\geq 1} j^2/2^j = 6$. Derive the actual ceiling upper bound $hu \leq 4M$, bound evaluation at $-t/(2L)$ and propagate errors using $p_i = p + 8hD_i + 8M(a+1)$. Those bounds do not follow from exact uniqueness alone.

A later test should distinguish this actual precision from the old doubled-budget proxy. It requires a separately prospective protocol and the precision proof before changing any campaign conclusion. The completed frozen panel was not rerun or relabelled, and no loss was attributed to host load.

Reproduction. With the pinned toolchain on PATH, from the repository root:

```
lake -d research/prime_common_scalar_20261108/lean build
lake -d research/prime_common_scalar_20261108/lean \
  env leanchecker --fresh --verbose HermiteUniqueness
```

The source gate receipts, exact commands, failed attempts and statement pins are included alongside the sources. No performance or benchmark row is promoted.